Program Pembangunan Olimpiad Matematik (Siri 1)

Algebra

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Contents

1 Algebraic Manipulations

1.1 Theory

Don't solve for the variables! Instead, try to directly find the value of the expression you want, by adding/subtracting/multiplying/squaring equations.

1.2 Problems

1. Given that $xy = 2$ and $x + y = 4$, find $x^2 + y^2$.

2. (CJR) If $143x - 77y = 451$, find the value of $299x - 161y$.

3. Given that $x - \frac{1}{x} = 4$, find $x^2 + \frac{1}{x^2}$.

4. (AMC 12) If x, y and z are positive numbers satisfying

$$x + \frac{1}{y} = 4,\ y + \frac{1}{z} = 1 \text{ and } z + \frac{1}{x} = \frac{7}{3},$$

then what is the value of xyz?

5. (CJR) Given that a, b, c are real numbers and

$$(b - 196a)^2 - 128(b - 14c)(c - 14a) = 0,$$

find the largest possible value of $\frac{b-14c}{c-14a}$.

2 Trivial Inequality

2.1 Theory

$$x^2 \geq 0 \text{ for all } x \in \mathbb{R}$$

2.2 Problems

1. Find x if $(x - 2025)^2 = 0$.
2. Find x, y, z if $22(x - 22)^{22} + 24(y - 24)^{24} + 26(z - 26)^{26} = 0$.
3. By completing the square, find the minimum value of
$$x^2 + 20x + 25.$$
4. (AMC 12) What is the least possible value of
$$(xy - 1)^2 + (x + y)^2$$
for real numbers x and y.
5. Find all ordered pairs of real numbers (x, y) such that
$$(4x^2 + 4x + 3)(y^2 - 6y + 13) = 8.$$
6. Prove that $a^2 + b^2 \geq 2ab$.
7. Find all solutions x, y, z of the equation
$$x^2 + 5y^2 + 10z^2 = 4xy + 6yx + 2z - 1.$$

3 AM-GM Inequality

3.1 Theory

For $a_1, a_2, ..., a_n \in \mathbb{R}^+$,

$$\frac{a_1 + a_2 + ... + a_n}{n} \geq \sqrt[n]{a_1 a_2 ... a_n}.$$

Equality occurs if and only if $a_1 = a_2 = ... = a_n$.

3.2 Problems

1. $a + \frac{1}{a} \geq 2$
2. $2(a^2 + b^2) \geq (a+b)^2 \geq 4ab$
3. $3(a^2 + b^2 + c^2) \geq (a+b+c)^2 \geq 3(ab+bc+ca)$
4. $a^2 + b^2 + c^2 \geq ab + bc + ca$
5. $a^3 + b^3 \geq ab(a+b)$
6. $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$
7. $(a+b)(b+c)(c+a) \geq 8abc$
8. $(a_1 + a_2 + ... + a_n)(\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}) \geq n^2$
9. $(a + \frac{1}{b})(b + \frac{1}{c})(c + \frac{1}{a}) \geq 8$
10. $(a^4 + b^4 + c^4)(ab + bc + ca + a^3 + b^3 + c^3) \geq 15(abc)^3$
11. $\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} \geq a + b + c + d$
12. $a_1 a_2 ... a_n = 1 \implies (a_1 + 1)(a_2 + 1)...(a_n + 1) \geq 2^n$
13. $abc = 1 \implies a + b + c \leq a^2 + b^2 + c^2$

4 Problems

1. (Austrian JMO 2023 P1) Let x, y, z be nonzero real numbers with
$$\frac{x+y}{z} = \frac{y+z}{x} = \frac{z+x}{y}.$$
Determine all possible values of
$$\frac{(x+y)(y+z)(z+x)}{xyz}.$$

2. (Al-Khwarizmi IJMO 2024 P6) Let a, b, c be distinct real numbers such that $a+b+c=0$ and $a^2-b=b^2-c=c^2-a$. Evaluate all the possible values of $ab+ac+bc$.

3. (Turkey JMO 2020 P1) Determine all real number pairs (x, y) that satisfy the equation $2x^2+y^2+7=2(x+1)(y+1)$.

4. (Greece JMO 2019 P1) Find all triplets of real numbers (x, y, z) that are solutions to the system of equations $x^2+y^2+25z^2=6xz+8yz$ and $3x^2+2y^2+z^2=240$

5. (Al-Khwarizmi IJMO 2024 P3) Find all $x, y, z \in \left(0, \frac{1}{2}\right)$ such that
$$\begin{cases} (3x^2+y^2)\sqrt{1-4z^2} \geq z; \\ (3y^2+z^2)\sqrt{1-4x^2} \geq x; \\ (3z^2+x^2)\sqrt{1-4y^2} \geq y. \end{cases}$$

6. (JBMO SL 2014 A2) Let a, b, c be positive real numbers such that $abc = \dfrac{1}{8}$. Prove the inequality:
$$a^2+b^2+c^2+a^2b^2+b^2c^2+c^2a^2 \geq \frac{15}{16}$$

7. (Greece JMO 2021 P1) If positive reals x, y are such that
$$2(x+y) = 1+xy,$$
find the minimum value of expression
$$A = x + \frac{1}{x} + y + \frac{1}{y}$$

8. (Turkey JMO 2022 P1) Suppose x, y, z are positive reals such that $x \leq 1$. Prove that
$$xy + y + 2z \geq 4\sqrt{xyz}$$